

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #6
Production Systems: Process Selection

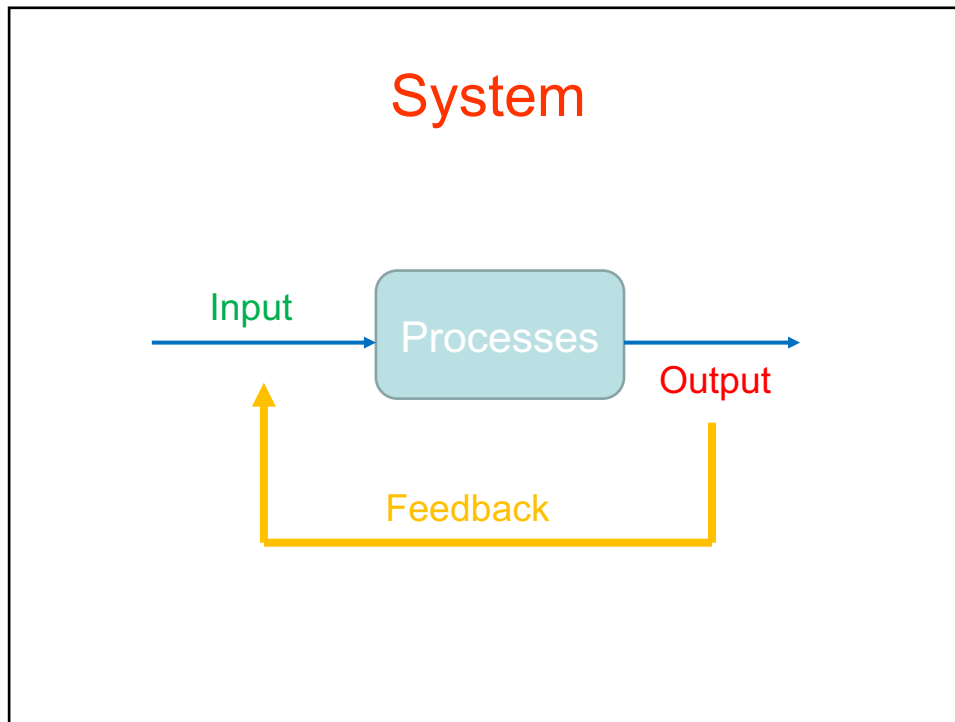
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Agenda of the Day

- Production system
- Processes
- Process selection
- Processes and technology

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What is a process?

- ❑ 2-minute exercise
- ❑ By yourself:
 - Write down what comes to your mind when you hear the word «Asia»
- ❑ Now, another 2-minutes: form groups of three by turning to your friends next to you:
 - Identify common themes and unique answers from each student
 - Discuss why some are common themes

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From Farm to Fork



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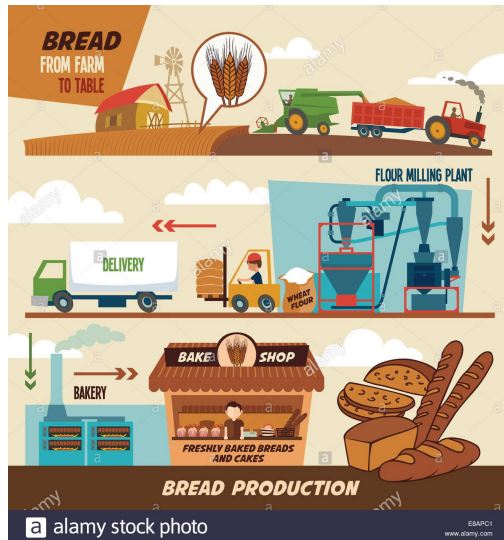
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Farm to table: five steps

- ❑ **Production,**
- ❑ **Processing,**
- ❑ **Distribution,**
- ❑ **Retailer and**
- ❑ **Consumer.**

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An example: Wheat



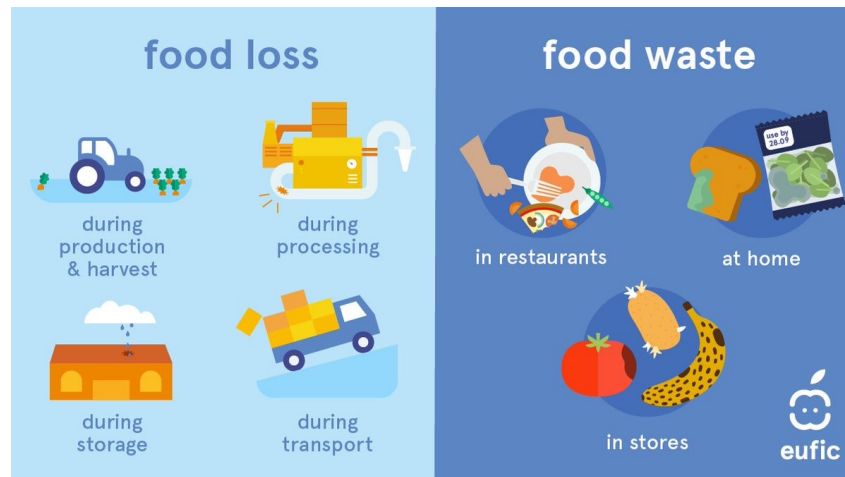
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A broader view



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A further step



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Production System

- A production system is anything that takes inputs and transforms them into outputs.
- Production system: the collection of people, equipment, **processes** and procedures organized to accomplish the operations of a company
 - Manufacturing: tangible
 - Service: usually not tangible

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What is a process?

- ❑ A series of actions performed systematically to fulfill a purpose.
- ❑ Through these actions certain things are transformed/changed/developed
- ❑ hopefully with an added value, and
- ❑ we will end up with a result/output

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Process Selection

- ❑ The ways organizations choose to produce or provide their goods and services.
- ❑ It involves choice of technology, type of processing, and so on.
- ❑ It influences
 - Capacity planning
 - Layout of facilities
 - Equipment
 - Design of work systems

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